Lecture on Grover's Algorithm 2/4/2025 G. del Corso

Given a set X of N elements $\to$ find $x \in X$ such that $f(x) = 1$

ex: find $x, y, z, w \in \mathbb{N}$: $0 < x, y, z, w \leq 10^6$

$$x^4 + y^4 + z^4 = w^4$$

check if $(x^4 + y^4 + z^4)^{1/4} \in \mathbb{N}$

the search space has size $O(10^{18})$

- Brute force algorithm $\to$ $O\left(\frac{10^{18}}{6}\right)$ trying all the possible x, y, z

$\downarrow$ in general $O(N)$ access to the "oracle"

$$i \to \boxed{f} \to f(i)$$

If X has t occurrences of elements such that $f(x) = 1$

$\Rightarrow$ $N - t$ access to oracle

- Randomized algorithm $O\left(\frac{N}{t}\right)$
- Quantum algorithm $O\left(\sqrt{\frac{N}{t}}\right)$ — optimal in the quantum framework

If t is large the quadratic speedup of Grover's alg. can be not so significant.

CAVEAT $\to$ (1) Access to data $<$ oracles $\cdot$ $U_O|i\rangle = x_i$ / QRAM model

(2) Cost of quantum operations respect to classical

if C is the cost of a quantum ops.
c is the cost of a classic ops.

$\boxed{t=1}$

$$C\sqrt{N} < cN \qquad \frac{C}{c} < \sqrt{N} \iff N > \left(\frac{C}{c}\right)^2$$

Denote by G the set of "good" solutions.

$$f(x) = \begin{cases} 0 & \text{if } x \notin G \\ 1 & \text{if } x \in G \end{cases}$$

ex: diophantine eq.

$$f(x,y,z) = \begin{cases} 0 & \text{if } (x^4+y^4+z^4)^{1/4} \notin \mathbb{N} \\ 1 & \text{if } (x^4+y^4+z^4)^{1/4} \in \mathbb{N} \end{cases}$$

The design of the quantum oracle is the trickiest part

$\to$ Reversible (unitary operator)

$\to$ implement easily unitary ops

WLG $\qquad X = \{|0\rangle, |1\rangle, \dots, |N-1\rangle\}$

$$f(i) = \begin{cases} 0 & \text{if } i \notin G \\ 1 & \text{if } i \in G \end{cases}$$

Grover Alg.

1. We start with an equal superposition of all the possible states

$$|\varphi\rangle = \frac{1}{\sqrt{N}} \sum_{i=0}^{N-1} |i\rangle = \frac{1}{\sqrt{N}} \begin{pmatrix} 1 \\ \vdots \\ 1 \\ \\ 1 \end{pmatrix}$$

goal is to move

$$|G\rangle = \frac{1}{\sqrt{t}} \sum_{i:f(i)=1} |i\rangle = \frac{1}{\sqrt{t}} \begin{pmatrix} 0 \\ \vdots \\ 1 \\ 0 \\ \vdots \\ 1 \\ 0 \\ \vdots \\ 1 \end{pmatrix}$$

$f(i)=1$

$$|\perp\rangle = \frac{1}{\sqrt{N-t}} \sum_{i:f(i)=0} |i\rangle = \frac{1}{\sqrt{N-t}} \begin{pmatrix} 1 \\ \vdots \\ 1 \\ 0 \\ 1 \\ \vdots \\ 0 \\ \vdots \\ 1 \end{pmatrix}$$

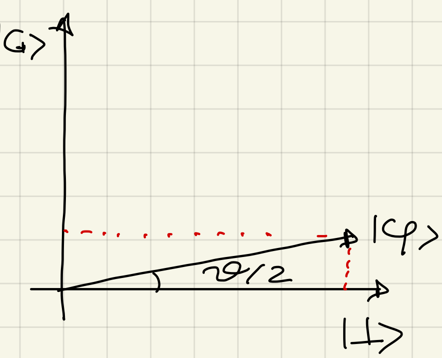
$f(i)=1$

"bad" states

$\langle \perp | G \rangle = 0$ Good and bad states are orthogonal

$$|\varphi\rangle = \boxed{\sqrt{\frac{t}{N}}}\,|G\rangle + \sqrt{\frac{N-t}{N}}\,|\perp\rangle$$

$$\sqrt{1 - \frac{t}{N}}$$



$$|\varphi\rangle = \boxed{\sin\frac{\theta}{2}}\,|G\rangle + \boxed{\cos\frac{\theta}{2}}\,|\perp\rangle$$

$$\sin\frac{\theta}{2} = \sqrt{\frac{t}{N}} \Rightarrow \theta = 2\arcsin\left(\sqrt{\frac{t}{N}}\right)$$

Repeat k times the following steps:

- Make a phase-query

$$|i\rangle \to (-1)^{f(i)}|i\rangle$$

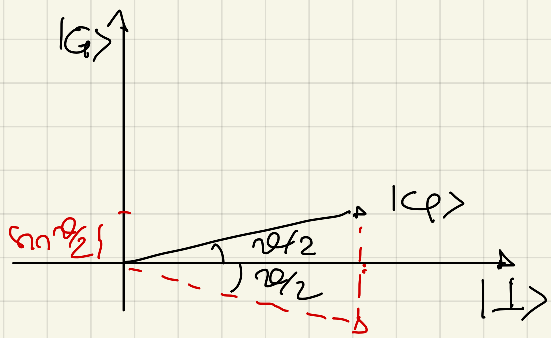
$f(i)=0 \Rightarrow (-1)^0 = 1$ nothing happens

$f(i)=1 \Rightarrow (-1)^1 = -1$ a phase is introduced

$$U_f = \begin{bmatrix} 1 & & & & & \\ & \ddots & & & & \\ & & -1 & & & \\ & & & 1 & & \\ & & & & \ddots & \\ & & & & & -1 \\ & & & & & & \ddots \\ & & & & & & & 1 \end{bmatrix} = \begin{bmatrix} (-1)^{f(0)} & & & \\ & (-1)^{f(1)} & & \\ & & \ddots & \\ & & & (-1)^{f(N-1)} \end{bmatrix}$$

$$U_f|\varphi\rangle = U_f\left(\sin\frac{\vartheta}{2}|G\rangle + \cos\frac{\vartheta}{2}|\perp\rangle\right) =$$

$$= -\sin\frac{\vartheta}{2}|G\rangle + \cos\frac{\vartheta}{2}|\perp\rangle$$



U_f is a reflection along $|\perp\rangle$

- Apply the diffuser

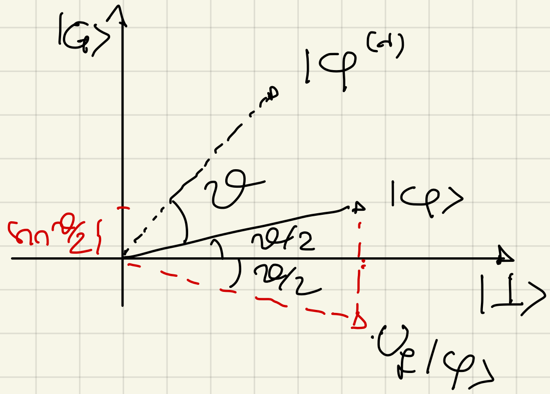
$$U_\varphi = 2|\varphi\rangle\langle\varphi| - I \qquad |\varphi\rangle = H^{\otimes n}|0\rangle^{\otimes n}$$

$$= H^{\otimes n}\underbrace{\left(2|0\rangle\langle 0| - I\right)}_{U_0}H^{\otimes n}$$

$U_0 \leftarrow$ reflection around the all zero state

$$U_0 = 2\begin{bmatrix}1\\0\\ \vdots\\ \vdots\\ 0\end{bmatrix}\begin{bmatrix}1 & \cdots & 0\end{bmatrix} - I = 2\begin{bmatrix}1 & & \\ & 0 & \\ & & \end{bmatrix} - I = \begin{bmatrix}1 & & & \\ & -1 & & \\ & & \ddots & \\ & & & -1\end{bmatrix}$$

U_φ reflects $U_f|\varphi\rangle$ around $|\varphi\rangle$



the composition of the two reflections $R = U_\varphi U_f$ is a rotation of ϑ

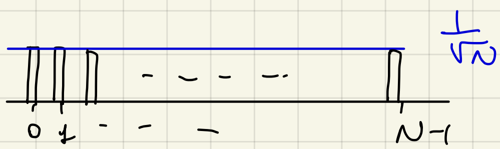
After k steps the angle is $k\vartheta + \frac{\vartheta}{2}$

I have to stop when $k\vartheta + \frac{\vartheta}{2} = \frac{\pi}{2}$

$$k = \frac{\pi}{2\vartheta} - \frac{1}{2}$$

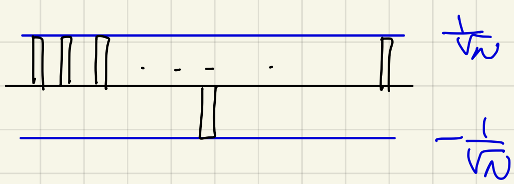
in general this value is not an integer

$|\varphi\rangle$



$t = 1$

$U_f|\varphi\rangle$



$U_\varphi U_f|\varphi\rangle$

After the application of the oracle

$$|z\rangle = U_f|\varphi^{(n)}\rangle = \sum_{i=0}^{N-1} \alpha_i |i\rangle \qquad \sum_{i=0}^{N-1} \alpha_i^2 = 1$$

$$U_\varphi |z\rangle = \left(2|\varphi\rangle\langle\varphi| - I\right)|z\rangle =$$

$$= 2|\varphi\rangle\langle\varphi|z\rangle - |z\rangle = 2\langle\varphi|z\rangle|\varphi\rangle - |z\rangle$$

$$= 2\sum_{i=0}^{N-1} \alpha_i \underbrace{\langle\varphi|i\rangle}_{=\frac{1}{\sqrt{N}}} |\varphi\rangle - \sum_{i=0}^{N-1} \alpha_i |i\rangle$$

$$\langle\varphi| \quad \left(\frac{1}{\sqrt{N}} \cdots \frac{1}{\sqrt{N}}\right) \qquad |i\rangle \quad \begin{pmatrix} 0 \\ \vdots \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \leftarrow i$$

$$= \left(\frac{2}{\sqrt{N}} \sum_{i=0}^{N-1} \alpha_i\right)|\varphi\rangle - |z\rangle$$

$$M = \text{mean}(\alpha_i) = \frac{1}{N}\sum_{i=0}^{N-1} \alpha_i \qquad = \frac{2\sqrt{N}}{N}\sum_{i=0}^{N-1}\alpha_i = 2M\sqrt{N}$$

$h=0$

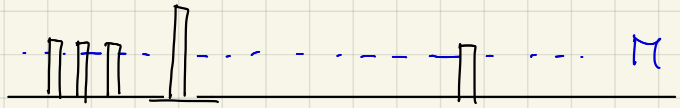
$$|z\rangle = U_f|\varphi\rangle = \frac{1}{\sqrt{N}}\sum_{i:f(i)=0} |i\rangle - \frac{1}{\sqrt{N}}\sum_{i:f(i)=1} |i\rangle$$

$$\alpha_i = \begin{cases} \frac{1}{\sqrt{N}} & \text{if } f(i)=0 \\ -\frac{1}{\sqrt{N}} & \text{if } f(i)=1 \end{cases}$$

$$M = \frac{1}{N}\left(\frac{N-t}{\sqrt{N}} - \frac{t}{\sqrt{N}}\right) = \frac{1}{N}\left(\frac{N-2t}{\sqrt{N}}\right)$$

$$|\varphi^{(1)}\rangle = U_\varphi|z\rangle = \left(2M + \frac{1}{\sqrt{N}}\right)\sum_{i:f(i)=1}|i\rangle + \left(2M - \frac{1}{\sqrt{N}}\right)\sum_{i:f(i)=0}|i\rangle$$

$U_\varphi U_f |\varphi\rangle$



$$k = \left\lceil \frac{\pi}{2\vartheta} - \frac{1}{2} \right\rfloor \leftarrow \text{rounding towards nearest integer}$$

<u>OSS</u>: if t is known we can "tune" ϑ to land on $|G\rangle$ exactly

since $\sin\frac{\vartheta}{2} = \sqrt{\frac{t}{N}}$ $\quad \frac{\vartheta}{2} = \arcsin\left(\sqrt{\frac{t}{N}}\right) \approx \sqrt{\frac{t}{N}}$

$$k = \left(\frac{\pi}{2} - \frac{\vartheta}{2}\right)\frac{1}{\vartheta} \approx \left(\frac{\pi}{2} - \sqrt{\frac{t}{N}}\right)\cdot\frac{1}{2}\sqrt{\frac{N}{t}} = \frac{\pi}{4}\sqrt{\frac{N}{t}} - \frac{1}{2}$$

$$k = \left\lfloor \frac{\pi}{4}\sqrt{\frac{N}{t}} \right\rfloor \quad \text{cost of Grover's Iterations}$$

$$\left(\frac{\pi}{4}\sqrt{\frac{N}{t}}\right)\left(\text{cost query} + \underbrace{\text{cost difuser}}_{O(\underbrace{n}_{\log N})\text{ gates}}\right)$$

<u>Theorem</u>: For large N, for $t=1$, if we perform $k = \left\lfloor \frac{\pi}{4}\sqrt{N} \right\rfloor$ Grover's iterations, the probability of measuring state $|G\rangle$ is

$$P(\text{measuring } |G\rangle) \geq 1 - \frac{1}{N}$$

Proof: $P(\text{measuring } |G\rangle) = \left| \langle G | (U_\varphi U_f)^k |\varphi\rangle \right|^2$

$$= \left| \langle G | \left(\sin\left(k\vartheta + \frac{\vartheta}{2}\right) |G\rangle + \cos\left(k\vartheta + \frac{\vartheta}{2}\right) |\perp\rangle \right) \right|^2$$

$$= \sin^2\left(k\vartheta + \frac{\vartheta}{2}\right) \underset{\sin x = \cos\left(\frac{\pi}{2} - x\right)}{=} \cos^2\left(\frac{\pi}{2} - \underbrace{\left(k\vartheta + \frac{\vartheta}{2}\right)}_{k = \frac{\pi}{4}\sqrt{N}}\right) = \cos^2\frac{\vartheta}{2} =$$

$$= 1 - \underbrace{\sin^2\frac{\vartheta}{2}}_{\left(\frac{1}{\sqrt{N}}\right)^2} = 1 - \frac{1}{N}$$